Data Cleaning Techniques

Road Traffic Accident (RTA) Dataset

Item	Detail
Source file	RTA Dataset.csv (preserved unchanged)
Size	12,316 rows x 32 columns
Scope	Data quality issues found in the source file and the cleaning technique defined for each. All techniques are applied on a working copy; the original file is never modified.

1. Purpose and guiding principles

This document records the data quality issues found in the full raw file: missing values, duplicates, inconsistent values, typos and data-type issues. For each issue it states the column, the number of affected records, the cleaning technique and the rationale. The techniques follow four principles: the original file is never modified; no record is deleted only because an optional field is blank; no replacement value is invented; every change is traceable to a documented reason.

2. Dataset overview

Check	Result	Meaning for the cleaning
Rows	12,316	Row count is preserved; no record is deleted.
Columns	32	Consistent with the project documentation (32 columns). No difference.
Exact duplicate rows	0	No duplicate-removal step is required. A second check excluding Time also found 0.
Blank cells	20,057	Real missingness across 16 columns.
Rows with at least one blank	9,427 (76.5%)	Deleting incomplete rows would remove most of the file, so records are kept.
Time column	No parse failures	Can be converted from text to Time type (see section 7).

3. Summary of issues and techniques

Every issue found, with the affected column, the number of records and the cleaning technique. Details and validation for each group are in sections 4 to 9.

#	Issue	Column(s)	Affected	Technique (and rationale)
1	Missing values (blank cells)	16 columns (section 4)	20,057 cells / 9,427 rows	Keep records; show blanks as 'Not Recorded' so they stay visible in slicers. Nothing is invented.
2	'na' in casualty fields	Casualty_class, Sex_of_casualty, Age_band_of_casualty, Casualty_severity	4,443 rows (36.1%)	Keep 'na' unchanged. It is a linked pattern, and these records still hold casualty counts, so it does not mean 'zero casualties'.
3	'Unknown' recorded values	Many columns (section 5)	e.g. 1,548 / 2,883 / 292	Keep as an explicit category. It is a recorded value, different from blank.
4	Leading spaces	Area_accident_occured	2,245	Trim the whole column. This also merges the two spellings of 'Recreational areas' (328 rows).
5	Two labels joined	Area_accident_occured	20	Keep as its own value with a clear name. It is not split, because the original meaning is not documented.
6	'?' instead of a dash	Type_of_vehicle	3,259	Replace '?' with '-' (e.g. Lorry (41-100Q)). Only the label changes.
7	Corrupted label	Pedestrian_movement	468	Restore to 'stationary'. The letters 'na' inside the word were replaced by 'Not a Pedestrian'.
8	Repeated text	Fitness_of_casuality	19	Map 'NormalNormal' to 'Normal'.

#	Issue	Column(s)	Affected	Technique (and rationale)
9	Time stored as text	Time	12,316 (2,808 before 10:00)	Convert to Time type. As text, hours before 10 have no leading zero and sort wrongly.
10	Case inconsistency	Lanes_or_Medians; Driving_experience	1,660 / 33	Standardise 'other' to 'Other' and 'unknown' to 'Unknown'.
11	Undocumented codes	Defect_of_vehicle (5, 7); Age_band_of_casualty (5); Casualty_severity (1, 2, 3)	112 / 244 / 7,873	Meanings are not assumed. Label as 'Code 5' and 'Code 7'; keep severity as text.
12	Similar-looking cause labels	Cause_of_accident	Not merged	Kept as separate categories; not merged (section 6).
13	Misspelled column names	4 columns (section 8)	4 headers	Keep source names; rename only in the reporting layer through a documented mapping.

4. Missing values: columns, counts and treatment

Rule for all 16 columns: keep the record and show the blank as 'Not Recorded'. A blank means the entry was not captured; replacing it with the most common value (mode) would invent data and distort the analysis. If a reliable value can be recovered from the source, it is used instead.

Column	Blank records	Rate	Technique	Rationale
Defect_of_vehicle	4,427	35.95%	Not Recorded	Highest gap. A blank may mean 'no defect' or 'not captured'; mode-fill would hide that.
Service_year_of_vehicle	3,928	31.89%	Not Recorded (separate from 'Unknown')	Blank and 'Unknown' (2,883) mean different things and must stay distinguishable.
Work_of_casuality	3,198	25.97%	Not Recorded (separate from 'Unknown')	Occupation cannot be inferred from other fields.
Fitness_of_casuality	2,635	21.39%	Not Recorded	Health status cannot be inferred.
Type_of_vehicle	950	7.71%	Not Recorded; recover from source if possible	Vehicle type drives several indicators, so recovery from the source is attempted first.
Types_of_Junction	887	7.20%	Not Recorded	Cannot be derived from other columns.
Driving_experience	829	6.73%	Not Recorded (separate from 'Unknown')	Experience cannot be inferred from age or other fields.
Educational_level	741	6.02%	Not Recorded (separate from 'Unknown')	Blank (741) and 'Unknown' (100) are kept apart.
Vehicle_driver_relation	579	4.70%	Not Recorded	Low missingness; keep visible.
Owner_of_vehicle	482	3.91%	Not Recorded	Low missingness; keep visible.
Lanes_or_Medians	385	3.13%	Not Recorded	Low missingness; keep visible.
Vehicle_movement	308	2.50%	Not Recorded	Low missingness; keep visible.
Area_accident_occured	239	1.94%	Not Recorded	Low missingness; keep visible. Trim first, then count blanks.
Road_surface_type	172	1.40%	Not Recorded	Low missingness; keep visible.
Type_of_collision	155	1.26%	Not Recorded	Low missingness; keep visible.
Road_allignment	142	1.15%	Not Recorded	Low missingness; keep visible.

Validation: after treatment, the 20,057 blank cells must read 'Not Recorded' and no unlabeled null should remain. Row count stays 12,316.

5. 'Unknown' and 'na' values

Blank, 'Unknown' and 'na' are three different things and are never merged. Blank is a missing entry. 'Unknown' is a value that was recorded. 'na' in the casualty fields is treated as no casualty detail recorded (see section 9).

Column	Value	Affected	Technique	Validation
Age_band_of_driver	Unknown	1,548	Keep. Do not infer age.	Count stays 1,548.

Column	Value	Affected	Technique	Validation
Sex_of_driver	Unknown	178	Keep. Do not guess sex.	Count stays 178.
Educational_level	Unknown	100	Keep; blanks become 'Not Recorded' so both meanings stay separate.	Both counts stay separate.
Vehicle_driver_relation	Unknown	14	Keep.	Count unchanged.
Driving_experience	unknown (lowercase)	33	Standardise to 'Unknown'.	Moves to the main 'Unknown' count; no row loss.
Service_year_of_vehicle	Unknown	2,883	Keep. Do not infer service age.	Unknown and blank counted separately.
Area_accident_occured	Unknown	22	Keep after trimming spaces.	No row loss.
Lanes_or_Medians	Unknown	57	Keep.	No row loss.
Types_of_Junction	Unknown	191	Keep.	No row loss.
Weather_conditions	Unknown	292	Keep.	No row loss.
Type_of_collision	Unknown	14	Keep.	No row loss.
Vehicle_movement	Unknown	88	Keep.	No row loss.
Work_of_casuality	Unknown	19	Keep; blanks stay a separate category.	No row loss.
Cause_of_accident	Unknown	25	Keep.	No row loss.
Casualty_class, Sex_of_casualty, Age_band_of_casualty, Casualty_severity	na	4,443 in each	Keep 'na' unchanged. Not deleted and not converted to 'Unknown'.	The same 4,443 rows stay synchronised across all four fields.

Important finding: the same 4,443 rows carry 'na' in all four casualty fields, so this is one structured pattern and not four separate typos. These rows also still have Number_of_casualties values from 1 to 8, so 'na' cannot mean 'zero casualties'.

6. Text quality and inconsistent categories

Column	Issue	Affected	Technique	Validation
Area_accident_occured	Leading spaces	2,245	Trim leading and trailing spaces in the whole column (all text columns should be trimmed as a general step).	Value counts before and after; no row change.
Area_accident_occured	'Recreational areas' written two ways (trimmed and untrimmed)	328 across both spellings	Trim first; the two spellings collapse into one category.	One 'Recreational areas' category remains.
Area_accident_occured	'Rural village areasOffice areas' (two labels joined)	20	Keep as its own value with a clear name, e.g. 'Rural village areas / Office areas'. It is not split because the original meaning is not documented.	Only these 20 rows change.
Fitness_of_casuality	'NormalNormal'	19	Map to 'Normal'.	'Normal' increases by exactly 19.
Pedestrian_movement	'statioNot a Pedestrianry'	468	Replace with 'stationary'. The letters 'na' inside the word were replaced by 'Not a Pedestrian'.	No corrupted label remains; total rows unchanged.
Type_of_vehicle	'?' in range labels (41?100Q, 11?40Q, 13?45 seats)	3,259	Replace '?' with '-'.	Counts per vehicle type stay the same; only labels change.
Driving_experience	unknown vs Unknown	33	Standardise to 'Unknown'.	No change in row count.
Lanes_or_Medians	other vs Other	1,660	Standardise to 'Other' (same meaning).	Category list becomes consistent.
Cause_of_accident	Similar labels: Overspeed (61) vs Driving at high speed (174); Turnover (78) vs Overturning (149)	Not merged	Kept as separate categories. They are not merged because their equivalence in meaning is not documented.	Category counts unchanged.

7. Data types and value checks

Field	Current situation	Technique	Validation
Time	Text. Hours before 10 have no leading zero (2,808 rows), so it sorts wrongly.	Convert to Time (h:mm:ss); the raw text is preserved in the original file.	0 parse failures; 12,316 valid times; hours between 0 and 23.
Number_of_vehicles_involved	Whole number, range 1 to 7.	Keep as Whole number.	Range check after cleaning; total 25,133.
Number_of_casualties	Whole number, range 1 to 8.	Keep as Whole number.	Range check after cleaning; total 19,067.
Casualty_severity	Text. Contains codes 1, 2, 3 (7,873 rows) mixed with 'na'.	Keep as text. Not treated as a numeric measure, because the code meanings are not documented.	Value counts unchanged.
Defect_of_vehicle	Text. Contains codes 5 and 7 (112 rows).	Label as 'Code 5' and 'Code 7'; meanings are not assumed.	Counts unchanged (112 records).
Age_band_of_casualty	Text. Contains code 5 (244 rows).	Label as 'Code 5'; meaning is not assumed.	Counts unchanged (112 records).
All other descriptive columns	Text.	Keep as text after trimming and normalisation. No numeric type forced on codes.	Review distinct values after cleaning.

8. Column-name issues

Four headers are misspelled in the source: **Area_accident_occured**, **Road_allignment**, **Work_of_casuality** and **Fitness_of_casuality**. The original names are kept in the source and cleaned data so everything stays traceable, and corrected display names are applied only in the reporting layer through a documented mapping table.

9. Handling of ambiguous values

Where the meaning of a value is not documented, the value is preserved and labelled rather than interpreted. This avoids introducing assumptions into the data.

Item	Records	Handling	Rationale
'na' in the four casualty fields	4,443	Kept unchanged; treated as no casualty detail recorded.	Linked pattern; records still hold casualty counts, so it does not mean zero casualties.
Codes 5 and 7 in Defect_of_vehicle; code 5 in Age_band_of_casualty	112 / 244	Labelled 'Code 5' and 'Code 7'.	Meaning is not documented; labelling keeps them visible without guessing.
Codes 1, 2, 3 in Casualty_severity	7,873	Kept as text.	No documented code dictionary; avoids treating codes as numeric values.
'Rural village areasOffice areas'	20	Kept as its own value with a clear name.	Two labels were joined; splitting would be a guess.
Similar labels in Cause_of_accident	Not merged	Kept as separate categories.	Equivalence in meaning is not documented; merging would change cause rankings.
Blank cells	20,057	Shown as 'Not Recorded' in all 16 columns.	Blanks stay visible in slicers and charts instead of disappearing.

10. Cleaning sequence and validation

Step	Action	Rationale
1	Create a working copy	The raw CSV is never edited.
2	Profile before changes (row count, nulls, distinct values)	Baseline for comparison.
3	Trim text in all text columns	Removes the 2,245 leading-space cases and hidden characters.

Step	Action	Rationale
4	Apply typo and label corrections	Pedestrian_movement, Fitness_of_casuality, Type_of_vehicle, Area_accident_occured.
5	Standardise case (unknown, other)	Same meaning, different spelling.
6	Handle blanks ('Not Recorded')	Keeps missing values visible without inventing data.
7	Leave 'na' and 'Unknown' unchanged	Different meaning from blank.
8	Fix data types (Time, whole numbers)	Correct sorting and calculations.
9	Re-profile and compare with the baseline	Detect any unwanted change.
10	Document every change (column, old value or rule, new value or rule, rows affected, reason)	Full traceability of every transformation.

Validation checks after cleaning:

- Row count stays 12,316 and column count stays 32.
- Accident_severity totals stay the same: Slight 10,415, Serious 1,743, Fatal 158.
- Totals stay the same: casualties 19,067 and vehicles 25,133.
- The 4,443 'na' rows and all 'Unknown' counts are unchanged (except the 33 'unknown' standardised to 'Unknown').
- Typo, case and space fixes move counts between labels but never change the total number of rows.
- No corrupted label remains ('statioNot a Pedestrianry', 'NormalNormal', '?' in vehicle types).

Summary: the techniques above fix inconsistent labels, typos and data types, and keep missing and unknown values visible, without deleting records or inventing values. Original meanings are preserved throughout.